

Dulmar Guud ee Nidaamka (System Overview)

DUGSI PRO 2026 waa nidaam dhammaystiran oo heer caalami ah oo loogu talagalay maamulka dugsiyada Islaamiga ah, macaahidda waxbarashada, iyo xarumaha xifdinta Qur'aanka Kariimka. Nidaamku wuxuu isku xiraa maamulka ardayda, macallimiinta, imtixaanaadka, jadwalka xiisadaha, maktabadda, hantida, iyo qayb weyn oo maaliyadeed (Invoicing, Payments, P&L, Budgets, Payroll).

Qaab-dhismeedka Farsamada (Technology Stack & Infrastructure)

Lakabka (Layer)	Tiknoolajiyadda (Technology)	Faahfaahinta Shaqada
Frontend UI	React 18+, TypeScript, Vite 6	Single Page Application (SPA) xawaare sare leh oo leh animations casri ah.
Styling & Icons	Tailwind CSS v4, Lucide React	Muuqaal qurux badan, nadiif ah, responsive ah oo mobile iyo desktop u diyaarsan.
Backend Server	Node.js, Express.js (Port 3000)	Server REST API leh oo maamula Auth, Multi-Tenancy, xisaabaadka, iyo Sync.
Primary Database	Supabase (PostgreSQL Cloud)	27 Tables leh xogta dugsiyada, ardayda, maaliyadda, iyo xiriirrada.
Resilient Fallback	Local File Storage (database.json)	Haddii internetku go'o ama Supabase la waayo, si toos ah offline ugu shaqeeya.
Export & Docs	PDF Generator, Excel/CSV Export	Rasiidhadda rasmiga ah, qaansheekooyinka, payslips, iyo warbixinada P&L.
Communication	Direct WhatsApp Integration	Hal-gujin loogu dirayo waalidka rasiidhadda, biilasha, iyo ogeysiisyada.

Tiirarka Waaweyn ee Nidaamka (Core Pillars)

Tiirka (Pillar)	Qaybaha ku Jira (Key Modules)	Faa'iidada Dugsiga
1. Academic & Student Core	Ardayda, Fasallada, Maadooyinka, Xaadirinta, Imtixaanaadka, Kaarka Natiijooyinka	Maamul dhamaystiran oo arday kasta taariikhdiisa iyo natiijooyinkiisa lagu hayo.
2. HR & Personnel	Macallimiinta, Shaqaalaha, Waalidiinta, Xaadirinta Shaqaalaha, Timetable	Isku xirka shaqaalaha, dhismaha jadwalka shaqada iyo xiriirka waalidiinta.
3. Operations & Campus	Qaabilaadda (Admissions), Maktabadda, Qalabka (Inventory), Ogeysiisyada	Habaynta ardayda cusub, hantida ma-guurtada ah iyo buugaagta maktabadda.
4. Institutional Finance	Fee Structures, Invoices, Payments, Expenses, Income, P&L, Budgets, Payroll	Xisaabaad xaqiiqo ah oo ka hortagaya wax isdaba-marin iyo khaladaadka lacagta.

Qaybaha Waxbarashada & Ardayda (Academic Modules Breakdown)

Qaybta (Module)	Awoodaha & Shaqooyinka Muhiimka ah	Habka Xogta & Shaashadda
Students Directory (Maamulka Ardayda)	- Diiwaangelin arday cusub oo wata sawir, taariikhda dhalashada, fasalka, iyo telefoonka waalidka. - Shaandhayn (Filter) fasal kasta ama xaalad (Active/Inactive). - Soo saaridda liiska ardayda oo Excel/CSV ah.	Shaashadda StudentsView & API: /api/students
360° Student Profile Modal	- Hal meel oo ardayga laga arko dhammaan taariikhdiisa: 1. Xogta Guud & Xiriirka Waalidka 2. Boqolleyda Xaadirinta (Attendance %) 3. Dhibcaha & Darajooyinka Imtixaanaadka 4. Hadhaaga Fiiga & Qaansheekooyinka u dhiman.	Pop-up Modal casri ah oo hal gujin ku furma
Attendance Tracking (Xaadirinta)	- Xaadirinta maalinlaha ah ee fasal kasta (Present, Absent, Late, Excused). - Kala saarista fadhiga Subaxa iyo Galabta (Session Type). - Tirakoob toos ah (Stats) oo muujinaya inta joogta iyo inta maqan.	Shaashadda AttendanceView & API: /api/attendance
Classes & Subjects (Fasallada & Maadooyinka)	- Dhismaha fasallada (Grade 1 ilaa Grade 12), Sections (A, B, C), iyo kuraasta (Capacity). - Qorista maadooyinka (Qur'aan, Tawxiid, Carabi, Xisaab, Saynis) iyo macallimiinta dhiga.	Shaashadaha ClassesView & SubjectsView
Exams & Report Cards (Imtixaanaadka)	- Galinta natiijooyinka imtixaanaadka (Marks Obtained vs Max Marks). - Xisaabinta tooska ah ee darajada (Grade: A+, A, B, C, D, Fail). - Soo saarista Kaarka Natiijada Ardayga (Report Card) oo daabacan.	Shaashadda ExamsView & API: /api/exam-scores

Jadwalka Xiisadaha & Qaabilaadda Cusub (Timetable & Admissions)

Qaybta	Sharaxaadda Shaqada	Xalka Farsamo
Timetable Schedule	Jadwal todobaadle ah (Sabti ilaa Jimce) oo fasal kasta u qeexaya saacadaha, maadooyinka, macallinka, iyo qolka. Waxay ka hortagtaa isku-dhaca macallimiinta ama qolalka.	TimetableScheduleView & dugsiga_timetable
Admissions & Intake	Maamulka codsiyada ardayda cusub ee doonaya inay is-diiwaangeliyaan. Maamulku wuxuu xaaladda ka dhigi karaa Pending, Approved, ama hal gujin ku rogi karaa arday rasmi ah (Enrolled).	AdmissionsView & dugsiga_admissions

Maamulka Shaqaalaha & Waalidiinta (Personnel & HR)

Doorka / Qaybta	Xogta La Maamulo	Xiriirka Nidaamka
Teachers (Macallimiinta)	Magaca, takhasuska (Specialization), heerka waxbarasho, fasallada iyo maadooyinka loo xilsaaray, mushaharka heshiiska ah, iyo taariikhda la shaqaaleeyay.	dugsiga_teachers & /api/teachers
Staff (Shaqaalaha Kale)	Xoghaynta, xisaabiyayaasha, ilaalada, nadaafadda, darawallada gaadiidka, waaxdooda (Department), iyo mushaharkooda bisha.	dugsiga_staff & /api/staff
Guardians (Waalidiinta)	Waalidka mas'uulka ka ah ardayda, telefoonka tooska ah, WhatsApp-ka, xiriirka qaraabada (Aabe, Hooyo, Abti), iyo ardayda u diiwaangashan.	dugsiga_guardians & /api/guardians
Staff Attendance	Xaadirinta maalinlaha ah ee macallimiinta iyo shaqaalaha (Present, Absent, Late, On Leave). Waxay saldhig u tahay xisaabinta mushaharka.	StaffAttendanceView & dugsiga_staff_attendance

Maktabadda, Hantida & Ogeysiisyada (Campus Assets & Circulation)

Qaybta	Ujeeddada & Hab-socodka	Natiijada Maamul
Maktabadda (Library Circulation)	Diiwaangelinta buugaagta (Kutubta, Tafiirka, Axaadiista, Maaddooyinka sayniska), tirada taalla, bixinta amaahda ardayda ama macallimiinta, iyo la socodka taariikhda soo celinta.	Ka hortagga lumitaanka buugaagta qaaliga ah ee dugsiga.
Hantida & Qalabka (Inventory)	Diiwaanka kuraasta, miisaska, kombiyuutarrada, qalabka cod-baahiyaha, qolalka ay yaallaan, xaaladdooda (Good, Fair, Needs Repair), iyo qiimaha iibsiga.	Ilaalinta hantida dugsiga iyo xisaabinta qiimaha qalabka yaalla.
Ogeysiisyada (Announcements)	Boodhka ogeysiisyada dugsiga ee lagu daabaco farriimaha degdegga ah. Waxaa loo kala saari karaa: Dhammaan, Macallimiinta kaliya, ama Waalidiinta kaliya.	Xog-gaarsiin degdeg ah oo xafiiska iyo qoysaska isku xirta.

Qaab-dhismeedka Xisaabaadka ee DUGSI PRO 2026

Qaybta Maaliyadda	Awoodaha & Xisaabinta	Amniga Xisaabeed
Fee Structures (Khidmadaha)	Dejinta qiimaha khidmadaha kala duwan: Lacagta bisha (Tuition), Diiwaangelinta, Gaadiidka, Imtixaanka, Buugaagta. Waxaa lagu xiri karaa fasallo gaar ah ama dugsiga oo dhan.	Qasab maaha in hal qiimo ardayda oo dhami bixiyaan.
Invoices (Qaansheekooyinka)	Jaridda qaansheekooyinka hal arday ama fasal dhan hal mar (Bulk Invoicing). Waxaa ku dhex jira line-items, qiimo-dhimis (Discounts), hadhaaga kaga dhiman, iyo daabacaadda PDF/WhatsApp.	Qaansheekadu waa biil sugaya lacag-bixin, kuma darsanto dakhliga ilaa la bixiyo.
Payments & Receipts (Rasiidhaha)	Qabashada lacagaha dhabta ah (Cash, Bank Transfer, EVC Plus, Zaad, Sahal). Waxay si toos ah hoos ugu dhigtaa hadhaaga qaansheekada waxayna soo saartaa rasiidh rasmi ah.	Lacagta qabashadeeda kaliya ayaa loo aqoonsadaa Dakhli soo galay (Realized Revenue).
Expenses (Kharashaadka)	Diiwaanka kharash kasta oo baxay (Ijaar, Koronto, Biyo, Dayactir, Agab). Waxaa la socda habka ogolaanshaha (Approval Workflow) iyo magaca qofkii ogolaaday.	Kharash aan la ogolaan lama dhexgeynayo xisaab-xirka P&L.
Profit & Loss (P&L Statement)	Warbixin maaliyadeed oo soo bandhigaysa: Dakhliga Guud - Kharashka Guud = Faa'iidada Saafiya ah (Net Profit). Waxay leedahay shaandhayn taariikheed iyo soo saarid Excel/PDF.	Ka hortagga in dakhliga ama kharashka laba jeer lagu celiyo (Double Counting Protection).
Budgets (Miisaaniyadda)	Qoondaynta miisaaniyadda sanadlaha ah ee waax kasta. Nidaamku wuxuu bixinayaa digniin guduudan (Over-budget warning) haddii kharashku dhaafo intii loo qoondeeyay.	Variance analysis xisaabisa farqiga u dhexeeya Planned vs Actual.
Payroll Engine (Mushahaarka)	Xisaabinta mushaharka: Basic Salary + Allowances - Deductions = Net Pay. Marka mushaharka la bixiyo, wuxuu toos ugu qormaayaa Expenses si uusan kharashku u lumin.	Daabacaadda xaashida mushaharka (Printable Payslip).

Ka-hortagga Khaladaadka Xisaabaadka (Double Counting Protection Rules)

- Invoices vs Payments: Qaansheekooyinka la jaray looma xisaabiyo dakhli dhab ah ilaa inta rasiidh lacag-bixin ah laga diiwaangeliyo.
- Payroll vs Expenses: Mushaharka marka la bixiyo waxaa loo qoraa hal kharash oo kaliya (Category = Salaries), loomana laba-laabo xisaabinta.
- Cash Flow Balance: Opening Cash + Payments Received + Other Income - Total Expenses = Closing Cash Balance.

Liiska Dhammaan 27-ka Miis ee Supabase (PostgreSQL Cloud Tables)

#	Magaca Miiska (Table)	Furaha (PK)	Qaybaha Muhiimka ah (Columns)	Doorka uu Qabto
1	dugsiga_users	id	email, password, role, verified	Akoonnada maamulka & shaqaalaha
2	dugsiga_students	id	school_id, full_name, class, gender, phone	Diiwaanka ardayda dugsiga
3	dugsiga_classes	id	school_id, class_name, teacher_name, room	Liiska fasallada dugsiga
4	dugsiga_subjects	id	school_id, subject_name, subject_code, class	Maadooyinka manhajka
5	dugsiga_exam_scores	id	school_id, student_id, marks, grade, term	Dhibcaha imtixaanaadka ardayda
6	dugsiga_attendance	Composite	school_id, date, student_id, status, session	Xaadirinta maalinlaha ah
7	dugsiga_fees	id	school_id, student_id, month, amount, paid	Diiwaankii hore ee fiiga
8	dugsiga_settings	Composite	school_id, key, value (JSONB)	Habaynta guud & lacagta dugsiga
9	dugsiga_teachers	id	teacher_id, name, phone, salary, classes	Diiwaanka macallimiinta
10	dugsiga_staff	id	employee_id, name, role, department, salary	Diiwaanka shaqaalaha dugsiga
11	dugsiga_guardians	id	name, phone, whatsapp, student_ids	Waalidiinta & xiriirkooda
12	dugsiga_staff_attendance	id	staff_id, staff_name, date, status	Xaadirinta shaqaalaha
13	dugsiga_timetable	id	class_name, teacher, subject, day, time	Jadwalka xiisadaha todobaadka
14	dugsiga_admissions	id	applicant_name, desired_class, status	Codsiyada ardayda cusub
15	dugsiga_announcements	id	title, message, audience, priority	Ogeysiisyada dugsiga
16	dugsiga_library_books	id	title, author, isbn, total_copies	Kataloogga buugaagta
17	dugsiga_library_loans	id	book_id, borrower_name, issue, due	Diiwaanka amaahashada buugaagta
18	dugsiga_inventory	id	item_name, category, quantity, condition	Hantida & qalabka dugsiga
19	dugsiga_documents	id	title, category, file_url, upload_date	Dokumentiyada & shahaadooyinka
20	dugsiga_notifications	id	title, message, channel, recipient	Farriimaha & ogeysiisyada
21	dugsiga_fee_structures	id	name, category, amount, class_name	Qaab-dhismeedka khidmadaha
22	dugsiga_invoices	id	invoice_number, student_id, total, balance	Qaansheekooyinka ardayda
23	dugsiga_payments	id	receipt_number, invoice_id, amount, method	Rasiidhada lacag-bixinta
24	dugsiga_expenses	id	category, description, amount, status	Kharashaadka baxay
25	dugsiga_income	id	category, description, amount, payer	Dakhliga kale ee dugsiga
26	dugsiga_budgets	id	category, planned_amount, actual_amount	Miisaaniyadda waaxaha
27	dugsiga_payroll	id	employee_name, gross, net_salary, status	Mushahaarka shaqaalaha

Heerarka Xuquuqda Isticmaalaha (Role-Based Access Control - RBAC)

Doorka (Role)	Awoodaha & Xeerarka Gelitaanka (Permissions)	Xayiraadaha (Restrictions)
Super Admin / Admin	Awood buuxda oo 100% ah dhammaan 29-ka qaybood: Ardayda, Macallimiinta, Xisaabaadka, Settings-ka, iyo Beddelka xogta.	Ma laha wax xaddidaad ah.
Accountant (Xisaabiye)	Gelitaanka buuxa ee Maaliyadda: Invoices, Payments, Expenses, Income, Budgets, Payroll, iyo P&L Reports.	Ma beddeli karo darajooyinka imtixaanaadka ardayda ama manhajka.
Teacher (Macallin)	Gelitaanka fasallada iyo maadooyinka loo xilsaaray, xaadirinta ardayda fasalkiisa, iyo galinta dhibcaha imtixaanka.	Ma arki karo xisaabaadka dugsiga, mushaharka dadka kale, ama settings-ka.
Staff / Receptionist	Qaabilaadda ardayda cusub (Admissions), xaadirinta shaqada, aragtida ogeysiisyada, iyo la socodka hantida.	Ma tirtiri karo xogta mana arki karo faa'iidada maaliyadda.

Amniga Xogta & Tilmaamaha Kormeeka (Security & Maintenance)

Qodobka Amniga	Habka uu Nidaamku u Fuliyo	Talooyinka Maamulka Dugsiga
Password Encryption	Furayaasha sirta ah waxaa lagu qariyaa crypto.scriptsSync iyadoo lagu darayo cryptographic salt.	U yeel isticmaalayaasha furayaal adag oo ka kooban xarfo iyo lambarro.
Database RLS Setting	Miisaska Supabase waxay ku jiraan UNRESTRICTED (Disable RLS) maadaama Node Server-ku maamulo amniga.	Ha Enable-gareynin RLS adigoo aan qorin SQL Policies, si aysan khaladaad u dhicin.
Multi-Tenancy (school_id)	Xog kasta waxay ku xiran tahay school_id gaar ah oo dugsiga u gaar ah.	Dugsiyo kala duwan ma arki karaan xogta midba midka kale.
Resilient Data Backup	Xogtu waxay ku xiran tahay Supabase Cloud, iyadoo sidoo kale database.json maxalli ah u kaydsan tahay.	Marmar ka qaado nuqul keyd ah (Download JSON Backup) shaashadda Settings.